Inference from sample statistics and margin of error

Problem-Solving and Data Analysis · Math · 28 questions

1 993a9eaf-5cb2-4e31-b138-e7b5002d37c5 · Easy

To estimate the proportion of a population that has a certain characteristic, a random sample was selected from the population. Based on the sample, it is estimated that the proportion of the population that has the characteristic is 0.49, with an associated margin of error of 0.04. Based on this estimate and margin of error, which of the following is the most appropriate conclusion about the proportion of the population that has the characteristic?

A. It is plausible that the proportion is between 0.45 and 0.53.

B. It is plausible that the proportion is less than 0.45.

C. The proportion is exactly 0.49.

D. It is plausible that the proportion is greater than 0.53.

2 e32ac8cc-1200-4ebd-9897-0c30b06a3bdf · Easy

A company designs and makes handbags. To estimate the mean weight of the handbags made by the company on a particular day, a sample of the handbags made by the company on that day was selected at random. Based on the sample, it is estimated that the mean weight of all handbags made by the company on that day is $27.8 \text{ ounces } \text{oz}$, with an associated margin of error of 0.02 oz. Based on this estimate and associated margin of error, which of the following is the most plausible conclusion?

A. The mean weight of all handbags made by the company on that day is between 27.78 oz and 27.82 oz.

B. The actual weights of all handbags made by the company on that day are between 27.78 oz and 27.82 oz.

C. The actual weights of all handbags from the sample are between 27.78 oz and 27.82 oz.

D. The mean weight of all handbags made by the company on that day is 27.8 oz.

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3 031666-DC · Easy

In a study, the data from a random sample of a population had a mean of 37, with an associated margin of error of 3. Which of the following is the most appropriate conclusion that can be made about the population mean?

A. It is less than 37.

B. It is greater than 37.

C. It is between 34 and 40.

D. It is less than 34 or greater than 40.

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4 026017-DC · Easy

A city has 50 city council members. A reporter polled a random sample of 20 city council members and found that 6 of those polled supported a specific bill. Based on the sample, which of the following is the best estimate of the number of city council members in the city who support the bill?

A. 6

B. 9

C. 15

D. 30

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5 029303-DC · Easy

At a large high school, 300 students were selected at random and were asked in a survey about a menu change in the school cafeteria. All 300 students completed the survey. It was estimated that 38% of the students were in support of a menu change, with a margin of error of 5.5%. Which of the following is the best interpretation of the survey results?

A. The percent of the students at the school who support a menu change is 38%.

B. The percent of the students at the school who support a menu change is greater than 38%.

C. Plausible values of the percent of the students at the school who support a menu change are between 32.5% and 43.5%.

D. Plausible values of the number of the students at the school who support a menu change are between 295 and 305.

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6 c2be4ad8-5127-48a8-8cd4-4a6d2c61e2c6 · Easy

An analyst collected data on the price of a carton of grape tomatoes at 30 locations selected at random in Utah. The mean price of a carton of grape tomatoes in Utah was estimated to be $\$ 4.23$, with an associated margin of error of $\$ 0.08$. Which of the following is a plausible statement about the mean price of a carton of grape tomatoes for all locations that sell this product in Utah?

A. It is between $\$ 4.15$ and $\$ 4.31$.

B. It is either less than $\$ 4.15$ or greater than $\$ 4.31$.

C. It is less than $\$ 4.15$.

D. It is greater than $\$ 4.31$.

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7 9dca7d0a-7c80-455e-807a-d6cf7cb2424d · Easy

Scott selected 20 employees at random from all 400 employees at a company. He found that 16 of the employees in this sample are enrolled in exactly three professional development courses this year. Based on Scott's findings, which of the following is the best estimate of the number of employees at the company who are enrolled in exactly three professional development courses this year?

A. 4

B. 320

C. 380

D. 384

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8 f12a8c8c-3eef-4c55-a726-f411ff595181 · Easy

There are 55 students in Spanish club. A sample of the Spanish club students was selected at random and asked whether they intend to enroll in a new study program. Of those surveyed, 20% responded that they intend to enroll in the study program. Based on this survey, which of the following is the best estimate of the total number of Spanish club students who intend to enroll in the study program?

A. 11

B. 20

C. 44

D. 55

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9 01550-DC · Easy

A random sample of 50 people from a town with a population of 14,878 were asked to name their favorite flavor of ice cream. If 7 people in the sample named chocolate as their favorite ice-cream flavor, about how many people in the town would be expected to name chocolate?

A. 350

B. 2,100

C. 7,500

D. 10,500

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10 08802-DC · Easy

A certain forest is 253 acres. To estimate the number of trees in the forest, a ranger randomly selects 5 different 1-acre parcels in the forest and determines the number of trees in each parcel. The numbers of trees in the sample acres are 51, 59, 45, 52, and 73. Based on the mean of the sample, which of the following ranges contains the best estimate for the number of trees in the entire forest?

A. 11,000 to 12,000

B. 12,500 to 13,500

C. 13,500 to 14,500

D. 18,000 to 19,000

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11 8a79b71f-d6aa-4062-b948-522622a8d42b · Medium

A researcher is designing a study to investigate the average number of hours participants in a training course spend on a particular activity per day. The researcher will report an estimated average number of hours participants in the training course spend on this activity per day with an associated margin of error. The researcher is considering using a random sample of either 115 or 230 participants from the training course. Which of the following would be the most likely effect of using the larger random sample compared to the smaller random sample?

A. The reported margin of error would be lower.

B. The reported margin of error would be higher.

C. The reported average number of hours would be lower.

D. The reported average number of hours would be higher.

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12 c6a93539-87f0-4e37-bc96-789723974cff · Medium

Researchers studied a sample of a species of fish from a certain location. The researchers found that the mean length of fish in the sample was 16.59 millimeters. Based on the sample mean and margin of error, the researchers estimated that the mean length of all fish of this species from this location was between 15.86 and 17.32 millimeters. What is the margin of error, in millimeters?

A. 0.73

B. 2.19

C. 15.86

D. 16.59

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13 1d0e4fb4-a269-44f0-b4ed-6434ddb20c94 · Medium

A company that produces socks wants to estimate the percent of the socks produced in a typical week that are defective. A random sample of 310 socks produced in a certain week were inspected. Based on the sample, it is estimated that 12% of all socks produced by the company in this week are defective, with an associated margin of error of 3.62%. Based on the estimate and associated margin of error, which of the following is the most appropriate conclusion about all socks produced by the company during this week?

A. 3.62% of the socks are defective.

B. It is plausible that between 8.38% and 15.62% of the socks are defective.

C. 12% of the socks are defective.

D. It is plausible that more than 15.62% of the socks are defective.

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14 070922-DC · Medium

A random sample of 400 town voters were asked if they plan to vote for Candidate A or Candidate B for mayor. The results were sorted by gender and are shown in the table below.

	Plan to vote for Candidate A	Plan to vote for Candidate B
Female	202	20
Male	34	144

The town has a total of 6,000 voters. Based on the table, what is the best estimate of the number of voters who plan to vote for Candidate A?

Student-produced response: _____

15 10642-DC · Medium

A study was done on the weights of different types of fish in a pond. A random sample of fish were caught and marked in order to ensure that none were weighed more than once. The sample contained 150 largemouth bass, of which 30% weighed more than 2 pounds. Which of the following conclusions is best supported by the sample data?

A. The majority of all fish in the pond weigh less than 2 pounds.

B. The average weight of all fish in the pond is approximately 2 pounds.

C. Approximately 30% of all fish in the pond weigh more than 2 pounds.

D. Approximately 30% of all largemouth bass in the pond weigh more than 2 pounds.

16 027102-DC · Medium

A store manager reviewed the receipts from 80 customers who were selected at random from all the customers who made purchases last Thursday. Of those selected, 20 receipts showed that the customer had purchased fruit. If 1,500 customers made purchases last Thursday, which of the following is the most appropriate conclusion?

A. Exactly 75 customers must have purchased fruit last Thursday.

B. Exactly 375 customers must have purchased fruit last Thursday.

C. The best estimate for the number of customers who purchased fruit last Thursday is 75.

D. The best estimate for the number of customers who purchased fruit last Thursday is 375.

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17 f2ee094c-7d77-427f-8375-dbbc8c6aa16b · Medium

Based on a random sample from a population, a researcher estimated that the mean value of a certain variable for the population is 20.5, with an associated margin of error of 1. Which of the following is the most appropriate conclusion?

A. It is plausible that the actual mean value of the variable for the population is between 19.5 and 21.5.

B. It is not possible that the mean value of the variable for the population is less than 19.5 or greater than 21.5.

C. Every value of the variable in the population is between 19.5 and 21.5.

D. The mean value of the variable for the population is 20.5.

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18 32ff895c-39e6-4e94-8a41-6783e35bf3ed · Medium

A company fills boxes with approximately 23 pounds of oranges. To test the accuracy of the filling process, 344 boxes of oranges were selected at random and weighed. Based on the sample, it is estimated that the average weight of all boxes of oranges filled by the company in an 8-hour period is 23.1 pounds, with an associated margin of error of 0.19 pounds. Which of the following is the best interpretation of this estimate?

A. Plausible values for the average weight of all boxes of oranges filled by the company are between 22.91 pounds and 23.29 pounds.

B. Plausible values for the average weight of all boxes of oranges filled by the company are less than 22.91 pounds or greater than 23.29 pounds.

C. The average weight of all boxes of oranges filled by the company is less than 23.01 pounds.

D. The average weight of all boxes of oranges filled by the company is greater than 23.01 pounds.

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19 c69d7c49-d5b4-474f-92d3-d2ef20e9813a · Medium

A sample consisting of 720 adults who own televisions was selected at random for a study. Based on the sample, it is estimated that 32% of all adults who own televisions use their televisions to watch nature shows, with an associated margin of error of 3.41%. Which of the following is the most plausible conclusion about all adults who own televisions?

A. More than 35.41% of all adults who own televisions use their televisions to watch nature shows.

B. Between 28.59% and 35.41% of all adults who own televisions use their televisions to watch nature shows.

C. Since the sample included adults who own televisions and not just those who use their televisions to watch nature shows, no conclusion can be made.

D. Since the sample did not include all the people who watch nature shows, no conclusion can be made.

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20 d0c73d6b-6fba-46e6-898b-9debc677ed82 · Medium

From a population of $50{,}000$ people, $1{,}000$ were chosen at random and surveyed about a proposed piece of legislation. Based on the survey, it is estimated that 35% of people in the population support the legislation, with an associated margin of error of 3%. Based on these results, which of the following is a plausible value for the total number of people in the population who support the proposed legislation?

A. 350

B. 650

C. $16{,}750$

D. $31{,}750$

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21 020263-DC · Medium

A park ranger asked a random sample of visitors how far they hiked during their visit. Based on the responses, the estimated mean was found to be 4.5 miles, with an associated margin of error of 0.5 miles. Which of the following is the best conclusion from these data?

A. It is likely that all visitors hiked between 4 and 5 miles.

B. It is likely that most visitors hiked exactly 4.5 miles.

C. It is not possible that any visitor hiked less than 3 miles.

D. It is plausible that the mean distance hiked for all visitors is between 4 and 5 miles.

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22 d3e4dff5-3775-4426-af52-2c2d27d2537f · Medium

A middle school faculty surveyed students to determine interest in an after-school chess club. Of the 100 students who were selected at random from all students in the school, 32% expressed interest in the club. Based on the sample mean and the margin of error, the faculty expects that between 24% and 40% of all students in the school would express interest in the club. What is the margin of error?

A. 0%

B. 8%

C. 16%

D. 32%

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23 029361-DC · Medium

A bag containing 10,000 beads of assorted colors is purchased from a craft store. To estimate the percent of red beads in the bag, a sample of beads is selected at random. The percent of red beads in the bag was estimated to be 15%, with an associated margin of error of 2%. If r is the actual number of red beads in the bag, which of the following is most plausible?

A. $r > 1{,}700$

B. $1{,}300 < r < 1{,}700$

C. $200 < r < 1{,}500$

D. $r < 1{,}300$

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24 07719-DC · Hard

In State X, Mr. Camp's eighth-grade class consisting of 26 students was surveyed and 34.6 percent of the students reported that they had at least two siblings. The average eighth-grade class size in the state is 26. If the students in Mr. Camp's class are representative of students in the state's eighth-grade classes and there are 1,800 eighth-grade classes in the state, which of the following best estimates the number of eighth-grade students in the state who have fewer than two siblings?

A. 16,200

B. 23,400

C. 30,600

D. 46,800

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25 ec66cd13-3613-4629-8a7c-c4dbfb4b53f7 · Hard

Poll Results

Angel Cruz	483
Terry Smith	320

The table shows the results of a poll. A total of 803 voters selected at random were asked which candidate they would vote for in the upcoming election. According to the poll, if 6,424 people vote in the election, by how many votes would Angel Cruz be expected to win?

A. 163

B. 1,304

C. 3,864

D. 5,621

Texting behavior	Talks on cell phone daily	Does not talk on cell phone daily	Total
Light	110	146	256
Medium	139	164	303
Heavy	166	74	240
Total	415	384	799

In a study of cell phone use, 799 randomly selected US teens were asked how often they talked on a cell phone and about their texting behavior. The data are summarized in the table above. Based on the data from the study, an estimate of the percent of US teens who are heavy texters is 30% and the associated margin of error is 3%. Which of the following is a correct statement based on the given margin of error?

A. Approximately 3% of the teens in the study who are classified as heavy texters are not really heavy texters.

B. It is not possible that the percent of all US teens who are heavy texters is less than 27%.

C. The percent of all US teens who are heavy texters is 33%.

D. It is doubtful that the percent of all US teens who are heavy texters is 35%.

27 029011-DC · Hard

Sample	Percent in favor	Margin of error
A	52%	4.2%
B	48%	1.6%

The results of two random samples of votes for a proposition are shown above. The samples were selected from the same population, and the margins of error were calculated using the same method. Which of the following is the most appropriate reason that the margin of error for sample A is greater than the margin of error for sample B?

A. Sample A had a smaller number of votes that could not be recorded.

B. Sample A had a higher percent of favorable responses.

C. Sample A had a larger sample size.

D. Sample A had a smaller sample size.

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28 992ca200-ff2c-49f6-87b0-189d4350f3cc · Hard

A researcher surveyed a random sample of $1{,}020$ people from a certain city to estimate the percentage of people who support a recently announced proposal. From the survey, the researcher estimated that 86% of people from the city support this proposal, with an associated margin of error of 2.13%. The researcher repeated the survey with a random sample of people from the city that was double the original sample size. Assuming the margins of error were calculated in the same way, which of the following is true about the margin of error associated with the estimate from the larger sample?

A. The margin of error is between 2.13% and 3.13%.

B. The margin of error is between 4.13% and 5.13%.

C. The margin of error is greater than 5.13%.

D. The margin of error is less than 2.13%.
